



Q1. Ans:
2^20 - 1 MB
So, 1 - 1/ 2^20 MB
So, 2^24 - (2^24/2^20) MB

$$= 2^4 \text{ MB}$$
$$= 16 \text{ MB}$$

Starting PA = 000000h
Last PA = FFFFFFFh

Q2. Ans.

Q3. Ans.

Q4. Ans:
Smallest:
i) (12345h - FFFFh) = 02346h

ii) Rounded down to nearest 10s = 02340h / 02350h

Seg = 0234/ 0235

iii) 0234h * 10h + FFFFh = 1233Fh
0235h * 10h + FFFFh = 1234Fh
0235h * 10h + FFF5h = 12345h

0235h = valid lowest segment

Highest:
i) (12345h - 0000h) = 12345h

ii) Rounded down to nearest 10s = 12340h / 12350h

Seg = 1234/ 1235

iii) 1235h * 10h + 0005h = 12355h
1234h * 10h + 0005h = 12345h

1234h = valid highest segment

Q5. Ans.
Q6. Ans. Logical addressing [Segment : offset]

Q7. Ans. 9A7B1h - 40FF0h = 597C1h; Here offset cannot be larger than 4 hex digits (invalid address)

Q8. Ans. PA = Segment * 10h + Offset
Segment = (PA - Offset)/10h
= (33330h - 0020h)/10h
= 3331h

Q9. Ans.
a) PA = base address/ segment * 10h + offset
= A4FBh * 10h + 012Bh
= A50DBh

b) PA = base address/ segment * 10h + 2nd lowest offset
= A4FBh * 10h + 0001h
= A4FB1h

c) PA = base address/ segment * 10h + 3rd highest offset
= A4FBh * 10h + FFFDh
= B4FADh

Q10. Ans. PA of Data segment = DS *10h + value of BX/ SI
PA of Code Segment = CS *10h + IP

Q11. Ans. 2^8 = 256. So the maximum 8-bit offset is FF.
2nd last address = 10000 + 00FE = 100FEh

Q12. Ans. PA of the string instruction = ES * 10h + BX
= 1100h * 10h + 2200h
= 13200h

Q13. Ans. 2^10 = 1024
= 1 K [After Rounding up]
= 1 K * 1 B
= 1KB

Q14. Ans.
Q15. Ans. Study from the slides